# Foundations of Operations Research

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# Chapter 3: Linear Programming (LP)

## Definition 1

A Linear Programming (LP) problem is an optimization problem:

$$\begin{array}{ll} \min & f(\underline{x}) \\ \text{s.t.} & \underline{x} \in X \subseteq \mathbb{R}^n \end{array} \tag{1}$$

where:

- the objective function $f : X \to \mathbb{R}$ is linear;
- the feasible region $X = \{\underline{x} \in \mathbb{R}^n : g_i(\underline{x}\ )\ r_i\ 0, i \in \{1, \ldots, m\}, r_i \in \{=, \geq, \leq\}\}$ with $g_i : \mathbb{R}^n \to \mathbb{R}$ linear, for $i = 1, \ldots, m$.

## Definition 2

$\underline{x}^* \in \mathbb{R}^n$ is an optimal solution of (1) if $f(\underline{x}^*) \leq f(\underline{x})$, $\forall \underline{x} \in X$.

A variety of decision-making problems can be formulated/approximated as LPs.

Often involve the optimal allocation of limited resources to different activities.

General form:

$$\begin{array}{ll} \min & z = c_1 x_1 + \cdots + c_n x_n \\ \text{s. t.} & a_{11} x_1 + \cdots + a_{1n} x_n \ (\geq, =, \leq) \ b_1 \\ & \vdots \\ & a_{m1} x_1 + \cdots + a_{mn} x_n \ (\geq, =, \leq) \ b_m \\ & x_1, \ldots, x_n \geq 0 \end{array}$$

Matrix notation

$$\begin{array}{ll} \min & z = \underline{c}^T \underline{x} \\ \text{s. t.} & A\underline{x} \geq \underline{b} \\ & \underline{x} \geq 0 \end{array} \quad ,$$

$$\begin{array}{ll} \min & z = [c_1 \ldots c_n] \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \\ \text{s. t.} & \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \geq \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix} \\ & \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \geq \underline{0} \end{array}$$

## Historical sketch

Jean B. Joseph Fourier
(1786–1830)

L. V. Kantorovitch
(1912–1986)

G. B. Dantzig
(1914–2005)

- 1825/26: Fourier presents a method to solve systems of linear inequalities and discusses LPs with 2-3 variables.
- 1939: Kantorovitch lays the foundations of LP and study the transportation problem (Nobel prize, 1975).
- 1947: Dantzig independently proposes LP and invents the Simplex algorithm.

**Example 1: Diet problem**

Given

$n$ aliments $j = 1, \dots, n$

$m$ nutrients (basic substances) $i = 1, \dots, m$

$a_{ij}$ amount of $i$-th nutrient contained in one unit of the $j$-th aliment

$b_i$ daily requirement of the $i$-th nutrient

$c_j$ cost of a unit of $j$-th aliment,

determine a diet that minimizes the total cost while satisfying all daily requirements.

**Model:**

Decision variables: $x_j$ = amount of $j$-th aliment in the diet, with $j = 1, \dots, n$

$$\begin{aligned} \min \quad & z = \sum_{j=1}^{n} c_j x_j \\ \text{s.t.} \quad & \sum_{j=1}^{n} a_{ij} x_j \geq b_i, \quad i = 1, \dots, m \\ & x_j \geq 0, \; j = 1, \dots, n \end{aligned}$$

**Example 2: Transportation problem (single product)**

Given

$m$ production plants $i = 1, \ldots, m$

$n$ clients $j = 1, \ldots, n$

$c_{ij}$ unit transportation cost from plant $i$ to client $j$

$p_i$ maximum supply (production capacity) of plant $i$

$d_j$ demand of client $j$

$q_{ij}$ maximum amount transportable from plant $i$ to client $j$

determine a transportation plan that minimizes the total costs while respecting plant capacities and client demands

**Assumption:** $\sum_{i=1}^{m} p_i \geq \sum_{j=1}^{n} d_j$

**Model:**

Decision variables: $x_{ij}$ = amount of product transported from $i$ to $j$, with $1 \leq i \leq m$ and $1 \leq j \leq n$

$$
\begin{aligned}
\min \quad & z = \sum_{i=1}^{m} \sum_{j=1}^{n} c_{ij} x_{ij} \\
\text{s.t.} \quad & \sum_{j=1}^{n} x_{ij} \leq p_i, \quad i = 1, \ldots, m && \text{(plant capacity)} \\
& \sum_{i=1}^{m} x_{ij} \geq d_j, \quad j = 1, \ldots, n && \text{(client demand)} \\
& 0 \leq x_{ij} \leq q_{ij}, \; i = 1, \ldots, m, \; j = 1, \ldots, n && \text{(transport. capacity)}
\end{aligned}
$$

**Example 3: Production planning problem**

Given

$m$ resources $i = 1, \dots, m$

$n$ products $j = 1, \dots, n$ which compete for resources

$c_j$ profit (selling price-cost) per unit of $j$-th product

$a_{ij}$ amount of $i$-th resource needed to produce one unit of $j$-th product

$b_i$ maximum available amount of i-th resource

determine a production plan that maximizes the total profit given the available resources.

**Model:** Decision variables: $x_j$ = amount of $j$-th product, with $j = 1, \dots, n$

$$\begin{aligned} \max \quad & z = \sum_{j=1}^{n} c_j x_j \\ \text{s.t.} \quad & \sum_{j=1}^{n} a_{ij} x_j \leq b_i, \quad i = 1, \dots, m \\ & x_j \geq 0, \; j = 1, \dots, n \end{aligned}$$

## Assumptions of LP models

1. Linearity of the objective function and constraints.

   Proportionality: Contribution of each variable = constant $\times$ variable
   Drawback: does not account for economies of scale.

   Additivity: Contribution of all variables = sum of single contributions
   Drawback: Competing products $\Rightarrow$ profits are not independent.

2. Divisibility: The variables can take fractional (rational) values.

3. Parameters are considered as constants which can be estimated with a sufficient degree of accuracy.

   More complex mathematical programs are needed to account for uncertainty.

LP "sensitivity analysis" allows to evaluate how sensitive an optimal solution is with respect to small changes in the parameter values (see end of Chapter 3).

# 3.1 Equivalent forms

General form:

$$
\begin{array}{lll}
\min(\max) & z = \underline{c}^T \underline{x} & \\
\text{s. t.} & A_1 \underline{x} \geq \underline{b_1} & \text{inequalities} \\
 & A_2 \underline{x} \leq \underline{b_2} & \text{inequalities} \\
 & A_3 \underline{x} = \underline{b_3} & \text{equations} \\
 & x_j \geq 0, \quad \text{for } j \in J \subseteq \{1, \ldots, n\} & \\
 & x_j \text{ free}, \quad \text{for } j \in \{1, \ldots, n\} \backslash J &
\end{array}
$$

## Definition 3 (Standard form)

$$
\begin{array}{lll}
\min & z = \underline{c}^T \underline{x} & \\
\text{s. t.} & A\underline{x} = \underline{b} & \text{only equality constraints and} \\
 & \underline{x} \geq \underline{0} & \text{all nonnegative variables}
\end{array}
$$

- The two forms are equivalent.
- Simple transformation rules allow to pass from one form to the other form.

**Warning:** The transformation may involve adding/deleting variables and/or constraints.

## Transformation rules

- $\max \underline{c}^T \underline{x} = -\min(-\underline{c}^T \underline{x})$
- $\underline{a}^T \underline{x} \leq b \Rightarrow \begin{cases} \underline{a}^T \underline{x} + s = b \\ s \geq 0 \end{cases}$ where $s$ is a slack variable.
- $\underline{a}^T \underline{x} \geq b \Rightarrow \begin{cases} \underline{a}^T \underline{x} - s = b \\ s \geq 0 \end{cases}$ where $s$ is a surplus variable.
- $x_j$ unrestricted in sign $\Rightarrow \begin{cases} x_j = x_j^+ - x_j^- \\ x_j^+, x_j^- \geq 0 \end{cases}$.

  After substituting $x_j$ with $x_j^+ - x_j^-$, we delete $x_j$ from the problem.

## Example

General form:

$$\begin{array}{ll} \max & f(x_1, x_2) = 2x_1 - 3x_2 \\ \text{s. t.} & 4x_1 - 7x_2 \leq 5 \\ & 6x_1 - 2x_2 \geq 4 \\ & x_1 \geq 0, \quad x_2 \in \mathbb{R} \end{array}$$

1. $x_2 = x_3 - x_4$, $x_3, x_4 \geq 0$:

$$\begin{array}{ll} \max & 2x_1 - 3x_3 + 3x_4 \\ \text{s. t.} & 4x_1 - 7x_3 + 7x_4 \leq 5 \\ & 6x_1 - 2x_3 + 2x_4 \geq 4 \\ & x_1, x_3, x_4 \geq 0 \end{array}$$

2. Introduce slack and surplus variables $x_5$ and $x_6$:

$$\begin{array}{ll} \max & 2x_1 - 3x_3 + 3x_4 \\ \text{s. t.} & 4x_1 - 7x_3 + 7x_4 + x_5 = 5 \\ & 6x_1 - 2x_3 + 2x_4 - x_6 = 4 \\ & x_1, x_3, x_4, x_5, x_6 \geq 0 \end{array}$$

3. Change the objective function sign:

$$\begin{array}{ll} \min & -2x_1 + 3x_3 - 3x_4 \\ \text{s. t.} & 4x_1 - 7x_3 + 7x_4 + x_5 = 5 \\ & 6x_1 - 2x_3 + 2x_4 - x_6 = 4 \\ & x_1, x_3, x_4, x_5, x_6 \geq 0 \end{array}$$

## Other straightforward transformations

- $a^T x \leq b \Leftrightarrow -a^T x \geq -b$
- $a^T x \geq b \Leftrightarrow -a^T x \leq -b$
- $a^T x = b \Leftrightarrow \begin{cases} a^T x \geq b \\ a^T x \leq b \end{cases} \Leftrightarrow \begin{cases} a^T x \geq b \\ -a^T x \geq -b \end{cases}$

# 3.2 Geometry of Linear Programming

**Example** Capital budgeting

Capital of 10 000€ and two possible investments A and B with, respectively, 4% and 6% expected return.

Determine a portfolio that maximizes the total expected return, while respecting the diversification constraints:

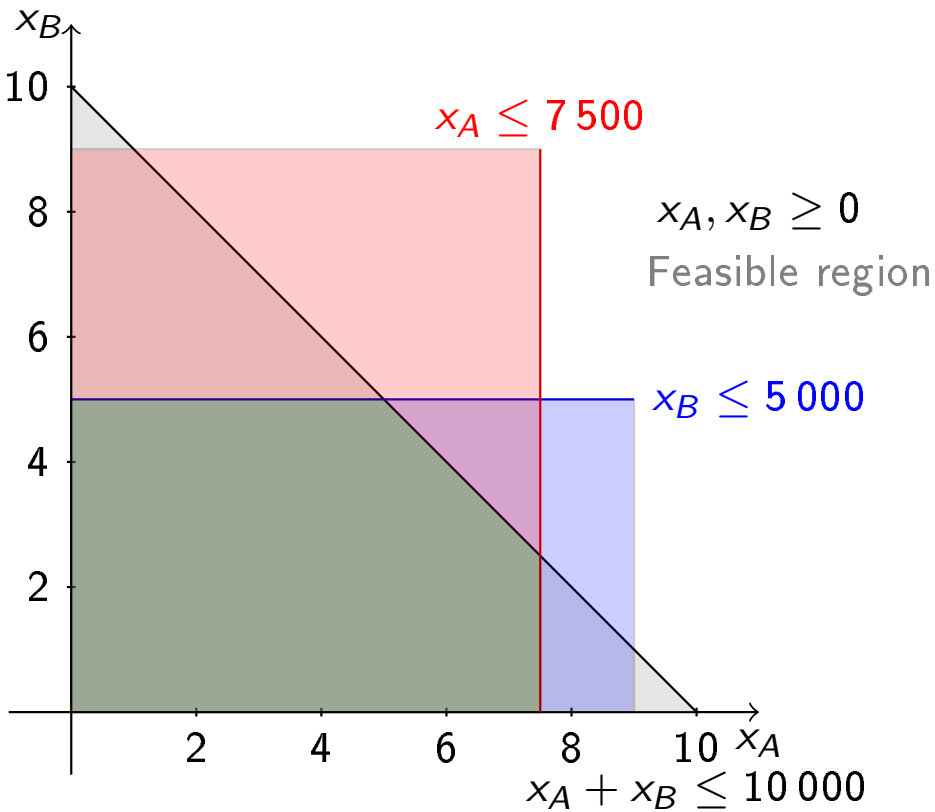
- at most 75% of the capital is invested in A,
- at most 50% of the capital is invested in B.

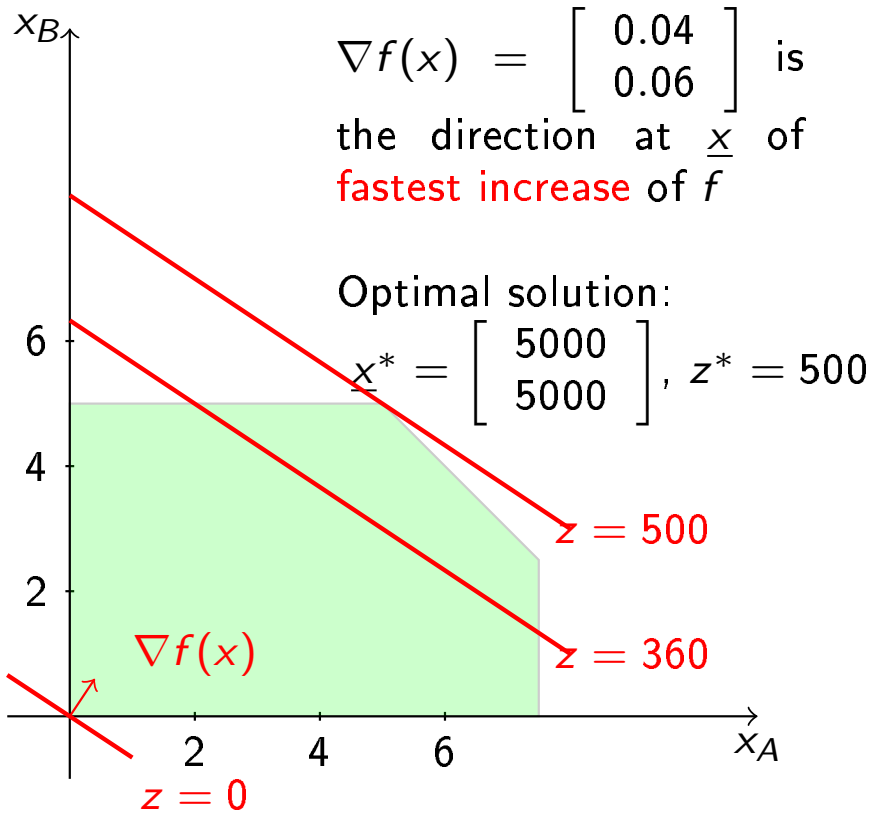
**Model:**

- $x_A$ = amount invested in A
- $x_B$ = amount invested in B

$$
\begin{array}{llrrcl}
\max & z = 0.04x_A & +0.06x_B & & & \\
\text{s.t.} & x_A & +x_B & \leq & 10\,000 \\
 & x_A & & \leq & 0.75 \cdot 10\,000 \\
 & & x_B & \leq & 0.50 \cdot 10\,000 \\
 & x_A, & x_B & \geq & 0
\end{array}
$$

# 3.2.1 Graphical solution




## Definition 4

A level curve of value $z$ of a function $f$ is the set of points in $\mathbb{R}^n$ where $f$ is constant and takes value $z$.

The level curves of a LP are lines: $0.04x_A + 0.06x_B = z$ ($z$ is a constant)

# 3.2.2 Vertices of the feasible region

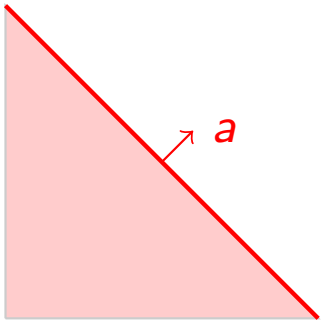
Consider a LP with inequality constraints (easier to visualize).

## Definition 5

- $H = \{\underline{x} \in \mathbb{R}^n : \underline{a}^T \underline{x} = b\}$ is a hyperplane
- $H^- = \{\underline{x} \in \mathbb{R}^n : \underline{a}^T \underline{x} \leq b\}$ is an affine half-space (half-plane in $\mathbb{R}^2$).

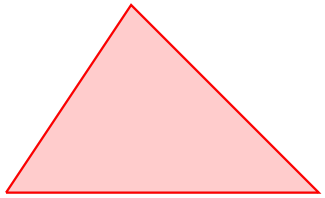
Each inequality constraint ($\underline{a}^T \underline{x} \leq b$) defines an affine half-space in the variable space:

$$H^- = \{\underline{x} \in \mathbb{R}^n : \underline{a}^T \underline{x} \leq b\}, \underline{a} \neq \underline{0}$$



## Definition 6

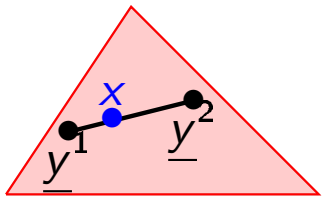
The feasible region $X$ of any LP is a polyhedron $P$. (Intersection of a finite number of half-spaces.)



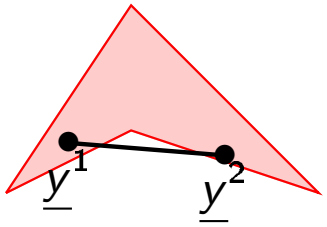
$P$ can be empty or unbounded

## Definition 7

A subset $S \subseteq \mathbb{R}^n$ is convex if for each pair $\underline{y}^1, \underline{y}^2 \in S$, $S$ contains the whole segment connecting $\underline{y}^1$ and $\underline{y}^2$.




convex




not convex

## Definition 8

The segment defined by $\underline{y}^1, \underline{y}^2 \in S$, defined by all the convex combinations of $\underline{y}^1$ and $\underline{y}^2$, is

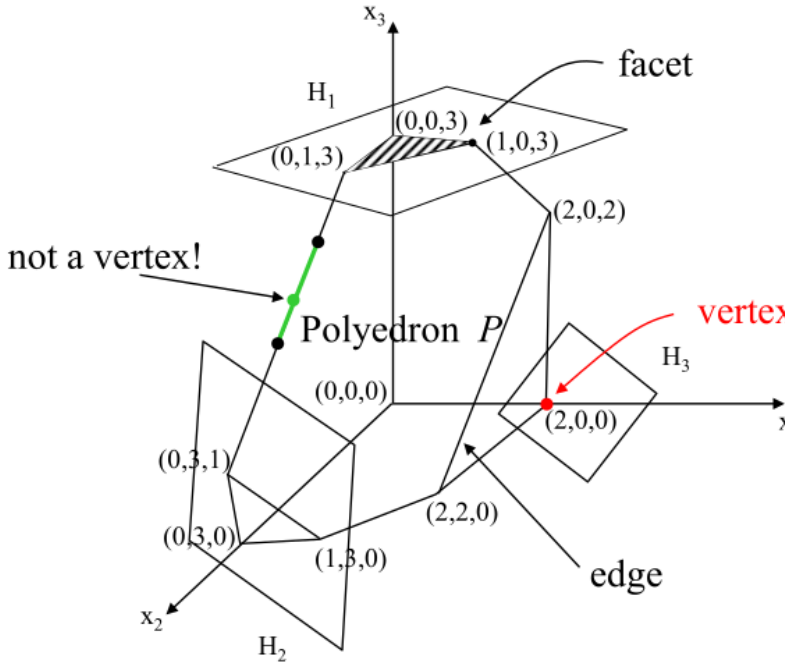
$$[\underline{y}^1, \underline{y}^2] = \{\underline{x} \in \mathbb{R}^n : \underline{x} = \alpha \underline{y}^1 + (1-\alpha)\underline{y}^2 \wedge \alpha \in [0,1]\}$$

## Property 1

*A polyhedron $P$ is a convex set of $\mathbb{R}^n$.*

Indeed:

- any half-space is convex;
- the intersection of a finite number of convex sets is also a convex set.




## Definition 9

A vertex of $P$ is a point of $P$ which cannot be expressed as a convex combination of two other distinct points of $P$.

Algebraically, $\underline{x}$ is a vertex of $P$ if and only if

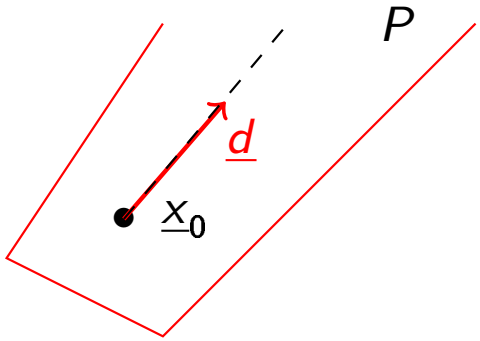
$$\underline{x} = \alpha \underline{y}^1 + (1-\alpha)\underline{y}^2,\ \alpha \in [0,1],\ \underline{y}^1, \underline{y}^2 \in P \Rightarrow \underline{x} = \underline{y}^1 \vee \underline{x} = \underline{y}^2$$

## Property 2

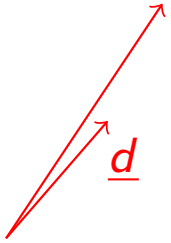
*A non-empty polyhedron* $P = \{\underline{x} \in \mathbb{R}^n : A\underline{x} = \underline{b}, \underline{x} \geq \underline{0}\}$ *(in standard form) or* $P = \{\underline{x} \in \mathbb{R}^n : A\underline{x} \geq \underline{b}, \underline{x} \geq \underline{0}\}$ *(in canonical form) has a finite number* ($\geq 1$) *of vertices.*

## Definition 10

Given a polyhedron $P$, a vector $\underline{d} \in \mathbb{R}^n$ with $\underline{d} \neq 0$ is an unbounded feasible direction of $P$ if, for every point $\underline{x}_0 \in P$, the "ray" $\{\underline{x} \in \mathbb{R}^n : \underline{x} = \underline{x}_0 + \lambda \underline{d}, \lambda \geq 0\}$ is contained in $P$.




unbounded feasible directions of $P$

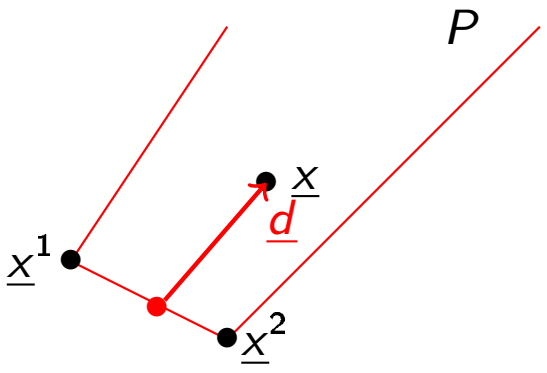


## Theorem 11 (Representation of polyhedra – Weyl-Minkowski)

*Every point $\underline{x}$ of a polyhedron $P$ can be expressed as a convex combination of its vertices $\underline{x}^1, \ldots, \underline{x}^k$ plus (if needed) an unbounded feasible direction $\underline{d}$ of $P$:*

$$\underline{x} = \alpha_1 \underline{x}^1 + \cdots + \alpha_k \underline{x}^k + \underline{d}$$

*where the multipliers $\alpha_i \geq 0$ satisfy $\alpha_1 + \cdots + \alpha_k = 1$.*

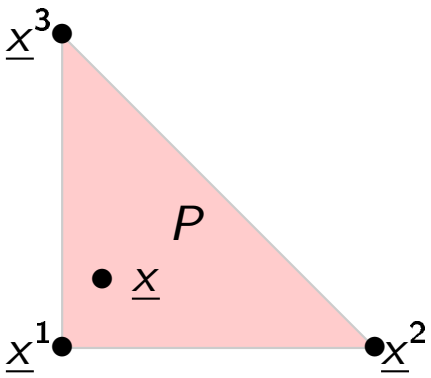



$$\forall \underline{x} \in [\underline{x}^1, \underline{x}^2], \underline{d} = \underline{0}$$

## Definition 12

A polytope is a bounded polyhedron, that is, it has the only unbounded feasible direction $\underline{d} = \underline{0}$.

**Consequence** Every point $\underline{x}$ of a polytope $P$ can be expressed as a convex combination of its vertices.




Then, $\underline{x} = \alpha_1 \underline{x}^1 + \alpha_2 \underline{x}^2 + \alpha_3 \underline{x}^3$, with $\alpha_i \geq 0$ and $\alpha_1 + \alpha_2 + \alpha_3 = 1$ ($\underline{d} = \underline{0}$).

## Theorem 13 (Fundamental theorem of Linear Programming)

*Consider a LP* $\min\{\underline{c}^T\underline{x} : \underline{x} \in P\}$, *where* $P \subseteq \mathbb{R}^n$ *is a non-empty polyhedron (in standard or canonical form). Then either there exists (at least) one optimal vertex or the value of the objective function is unbounded below on* $P$.

## Proof.

- Case 1: $P$ has an unbounded feasible direction $\underline{d}$ such that $\underline{c}^T\underline{d} < 0$.

  $P$ is unbounded and the values $z = \underline{c}^T\underline{x}$ tend to $-\infty$ along the direction $\underline{d}$.

- Case 2: $P$ has no unbounded feasible direction $\underline{d}$ such that $\underline{c}^T\underline{d} < 0$, that is, for all of them we have $\underline{c}^T\underline{d} \geq 0$.

  Any point of $P$ can be expressed as:

$$\underline{x} = \sum_{i=1}^{k} \alpha_i \underline{x}^i + \underline{d},$$

  where $\underline{x}^1, \ldots, \underline{x}^k$ are the vertices of $P$, $\alpha_i \geq 0$ with $\alpha_1 + \cdots + \alpha_k = 1$, and $\underline{d} = \underline{0}$, or $\underline{d}$ is an unbounded feasible direction.

...

□

## Proof.

- Case 2 (cont.): For any $\underline{x} \in P$, we have $\underline{d} = \underline{0}$ or $\underline{c}^T \underline{d} \geq 0$ and hence

$$\underline{c}^T \underline{x} = \underline{c}^T \left( \sum_{i=1}^{k} \alpha_i \underline{x}^i + \underline{d} \right) = \sum_{i=1}^{k} \alpha_i \underline{c}^T \underline{x}^i + \underline{c}^T \underline{d} \geq \min_{i=1,\ldots,k} \{\underline{c}^T \underline{x}^i\}$$

since $\alpha_i \geq 0$, for any $i$, and $\sum_{i=1}^{k} \alpha_i = 1$.

□

Although the variables are fractional, LPs can be viewed as combinatorial problems.

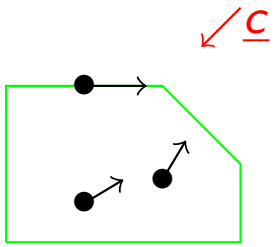
We "only" need to examine the vertices of the polyhedron of the feasible solutions!

The number of vertices is finite but often exponential.

The graphical method is only applicable for $n \leq 3$.

## Geometrically

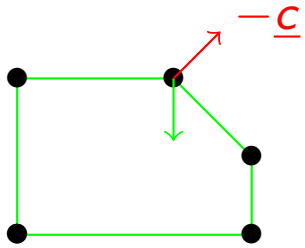
An interior point $\underline{x} \in P$ cannot be an optimal solution:




$\underline{c}$ = direction of fastest increase in $z$ (constant gradient)

$\exists$ always an improving direction

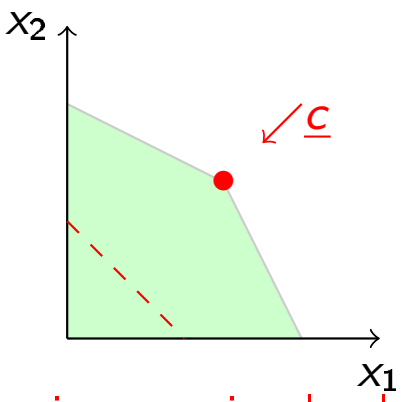
In an optimal vertex all feasible directions (for a sufficiently small step) are "worsening" directions:



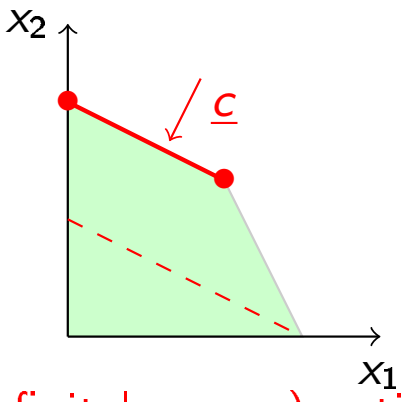

• vertices
improving directions
feasible directions

### 3.2.3 Four types of Linear Programs

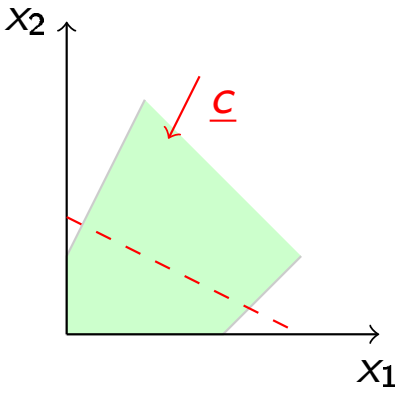
**Note:** Since $\min \underline{c}^T \underline{x}$, better solutions are found by moving along $-\underline{c}$.




A unique optimal solution

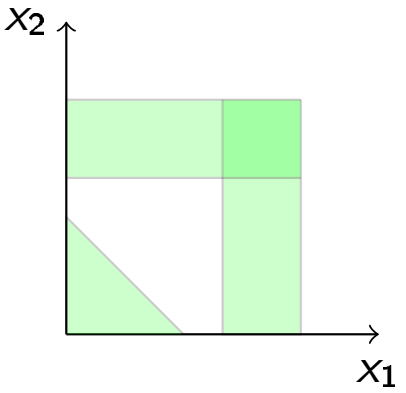



Multiple (infinitely many) optimal solutions




Unbounded polyhedron and unlimited objective function value

Unbounded LP

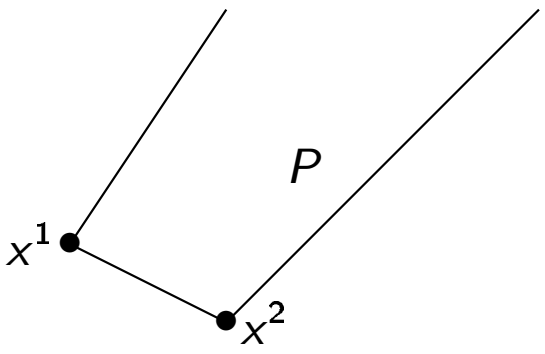



Empty polyhedron (no feasible solution)

Infeasible LP

# 3.3 Basic feasible solutions and vertices of polyhedra

Due to the LP fundamental theorem, to solve any LP it suffices to consider the vertices (finitely many) of the polyhedron P of the feasible solutions.

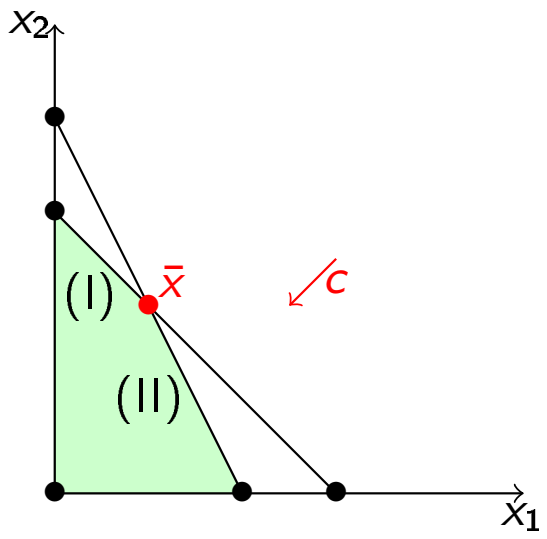



Since the geometrical definition of vertex cannot be exploited algorithmically, we need an algebraic characterization.

Which are the vertices of $P = \{\underline{x} \in \mathbb{R}^n : A\underline{x} \leq \underline{b}, \underline{x} \geq \underline{0}\}$ with only inequalities?

**Example**

$$\begin{array}{lrcll} \min & -x_1 - 3x_2 & & & \\ \text{s. t.} & x_1 + x_2 & \leq & 6 & \text{(I)} \\ & 2x_1 + x_2 & \leq & 8 & \text{(II)} \\ & x_1, x_2 & \geq & 0 & \end{array}$$




A vertex corresponds to the intersection of the hyperplanes associated to $n$ inequalities.

Vertex $\underline{\bar{x}}$ is the intersection of the hyperplanes of (I) and (II), i.e., the solution of the system:

$$\begin{cases} x_1 + x_2 = 6 \\ 2x_1 + x_2 = 8 \end{cases}$$

What about the vertices of polyhedra expressed in standard form?

$$P = \{\underline{x} \in \mathbb{R}^n \ : A\underline{x} = \underline{b}, \underline{x} \geq \underline{0}\}$$

We want to solve LPs in standard form. However, these are easier to describe if we start from

$$P = \{\underline{x} \in \mathbb{R}^n : \ A\underline{x} \leq \underline{b}, \underline{x} \geq \underline{0}\},$$

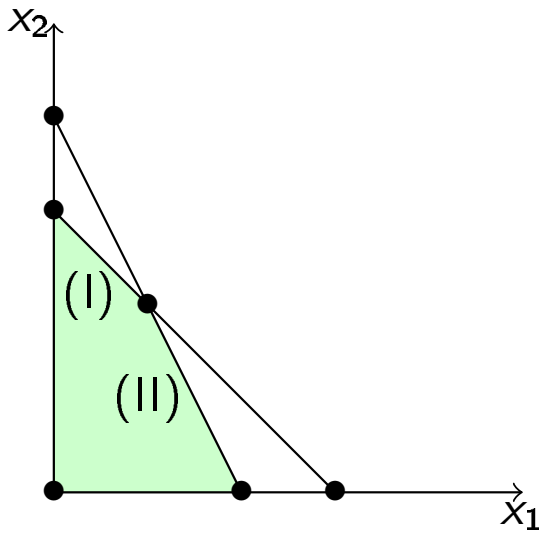
transform it into standard form

$$P' = \{\underline{x} \in \mathbb{R}^n : \ A\underline{x} + \underline{s} = \underline{b}, \underline{x}, \underline{s} \geq \underline{0}\}$$

and rename: $A := [A|I]$ and $\underline{x} := (\underline{x}^T|\underline{s}^T)$, where $A$ has $m$ rows.

**Example**

$$
(P)\quad \begin{array}{rcl} x_1 + x_2 & \leq & 6 \\ 2x_1 + x_2 & \leq & 8 \\ x_1, x_2 & \geq & 0 \end{array} \quad \Rightarrow (P')\quad \begin{array}{rcl} x_1 + x_2 + s_1 & = & 6 \\ 2x_1 + x_2 + s_2 & = & 8 \\ x_1, x_2, s_1, s_2 & \geq & 0 \end{array}
$$




Taking the intersection of the lines associated to (I) and (II) in $P$, amounts in $P'$ to let $s_1 = s_2 = 0$.

**Notes:**

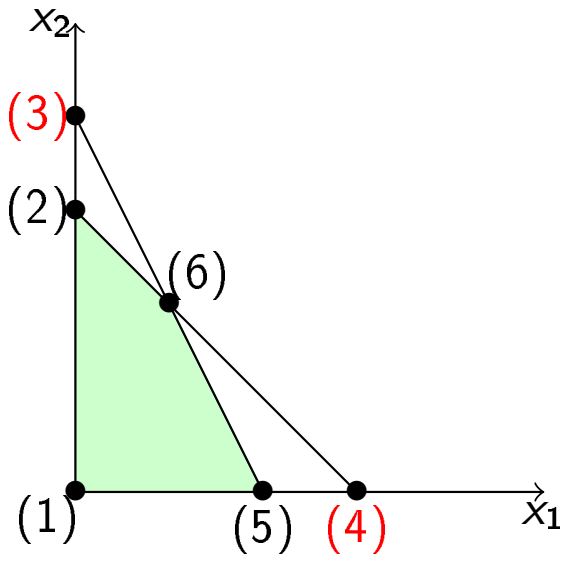
- Every constraint in $P$ corresponds to a slack variable in $P'$. When $s_i = 0$ the constraint is satisfied with "=".

  E.g. $s_1 = 0 \rightarrow x_1 + x_2 = 6$.
- Any vertex of $P$ is the intersection of the hyperplanes associated to $n$ inequalities. Equivalent to setting the corresponding variables in $P'$ to 0.

**Example** (continued)

Compute all the intersections

$$\begin{aligned} x_1 + x_2 + s_1 &= 6 \\ 2x_1 + x_2 + s_2 &= 8 \\ x_1, x_2, s_1, s_2 &\geq 0 \end{aligned}$$




(1) $x_1 = 0, x_2 = 0 \Rightarrow s_1 = 6, s_2 = 8$

(2) $x_1 = 0, s_1 = 0 \Rightarrow x_2 = 6, s_2 = 2$

(3) $x_1 = 0, s_2 = 0 \Rightarrow x_2 = 8, s_1 = -2$

(4) $x_2 = 0, s_1 = 0 \Rightarrow x_1 = 6, s_2 = -4$

(5) $x_2 = 0, s_2 = 0 \Rightarrow x_1 = 4, s_1 = 2$

(6) $s_1 = 0, s_2 = 0 \Rightarrow x_1 = 2, x_2 = 4$

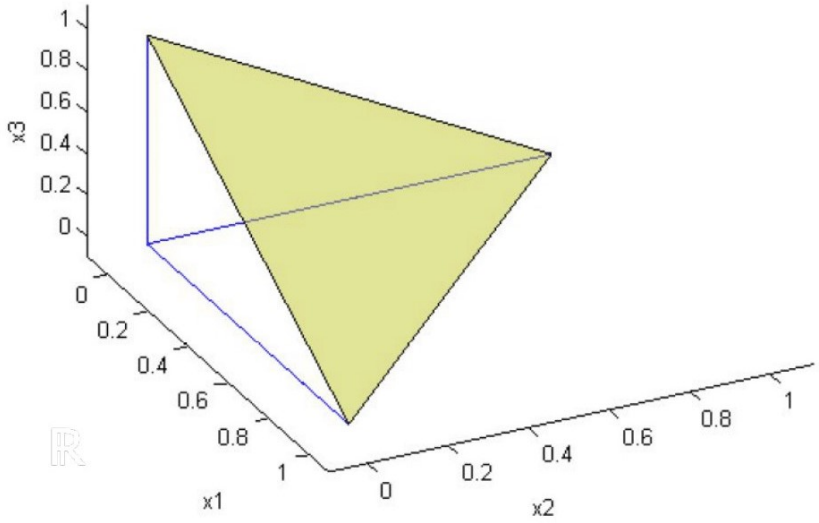
The intersections where some $x_j$ or $s_i$ are $< 0$ yield infeasible solutions.

Which are the vertices of a polyhedron in standard form?

**Example**

$$P = \{\underline{x} \in \mathbb{R}^3 : x_1 + x_2 + x_3 = 1, \underline{x} \geq \underline{0}\}$$

with $n = 3$ and $m = 1$.




## Property 3

*For any polyhedron $P = \{\underline{x} \in \mathbb{R}^n : A\underline{x} = \underline{b}, \underline{x} \geq \underline{0}\}$:*

1. *the facets (edges in $\mathbb{R}^2$) are obtained by setting one variable to 0,*
2. *the vertices are obtained by setting n-m variables to 0.*

In the above example: $3 - 1 = 2$ variables set to 0 for vertices.

## Algebraic characterization of the vertices

Consider any $P = \{\underline{x} \in \mathbb{R}^n : A\underline{x} = \underline{b}, \underline{x} \geq \underline{0}\}$ in standard form.

**Assumption:** $A \in \mathbb{R}^{m \times n}$ is such that $m \leq n$ of rank $m$ ($A$ is of full rank).

Equivalent to assume that there are no "redundant" constraints.

**Example**

$$\begin{array}{rrrcl} 2x_1 & +x_2 & +x_3 & = & 2 \text{ (I)} \\ x_1 & +x_2 & & = & 1 \text{ (II)} \\ x_1 & & +x_3 & = & 1 \text{ (III)} \\ x_1, & x_2, & x_3 & \geq & 0 \end{array}$$

Since (I) $\equiv$ (II) + (III), then (I) can be dropped.

- If $m = n$, there is a unique solution of $A\underline{x} = \underline{b}$ ($\underline{x} = A^{-1}\underline{b}$).
- If $m < n$, there are $\infty$ solutions of $A\underline{x} = \underline{b}$. The system has $n - m$ degrees of freedom, $n - m$ variables can be fixed arbitrarily. By fixing them to 0, we get a vertex.

$$P = \{\underline{x} \in \mathbb{R}^n : A\underline{x} = \underline{b}, \underline{x} \geq \underline{0}\}$$

with $n$ variables, $m$ constraints, and $A \in \mathbb{R}^{m \times n}$.

## Definition 14

A basis of such a matrix $A$ is a subset of $m$ columns of $A$ that are linearly independent and form an $m \times m$ non singular matrix $B$.

$$A = [\overbrace{B}^{m} \mid \overbrace{N}^{n-m}]$$

First permute the columns of $A$, then partition $A$ into $[B|N]$

Let $\underline{x} = [\ \overbrace{\underline{x}_B^T}^{m \text{ components}}\ |\ \overbrace{\underline{x}_N^T}^{n-m \text{ components}}\ ]^T$. Any system $A\underline{x} = \underline{b}$ can be written as $B\underline{x}_B + N\underline{x}_N = \underline{b}$. Since $B$ is non singular, for any set of values for $\underline{x}_N$ we have

$$\underline{x}_B = B^{-1}\underline{b} - B^{-1}N\underline{x}_N$$

## Definition 15

- A basic solution is a solution obtained by setting $\underline{x}_N = \underline{0}$ and, consequently, letting $\underline{x}_B = B^{-1}\underline{b}$.
- A basic solution with $\underline{x}_B \geq \underline{0}$ is a basic feasible solution (BFS).
- The variables in $\underline{x}_B$ are the basic variables, those in $\underline{x}_N$ are the non basic variables.

**Note:** By construction $(\underline{x}_B^T, \underline{x}_N^T)$ together satisfy $A\underline{x} = \underline{b}$.

## Theorem 16

*$\underline{x} \in \mathbb{R}^n$ is a basic feasible solution if and only if $\underline{x}$ is a vertex of $P = \{\underline{x} \in \mathbb{R}^n : A\underline{x} = \underline{b}, \underline{x} \geq \underline{0}\}$.*

**Example**

$$
\begin{array}{ll}
\min & 2x_1+x_2+5x_3 \\
\text{s. t.} & \begin{array}{rl}
x_1+x_2+x_3+x_4 & = 4 \\
x_1 \qquad\qquad +x_5 & = 2 \\
x_3 \qquad\quad +x_6 & = 3 \\
3x_2+x_3 \qquad\qquad +x_7 & = 6 \\
x_1,\ x_2,\ x_3,\ x_4,\ x_5,\ x_6,\ x_7 & \geq 0
\end{array}
\end{array}
\quad A = \begin{bmatrix} 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 3 & 1 & 0 & 0 & 0 & 1 \end{bmatrix} \quad b = \begin{bmatrix} 4 \\ 2 \\ 3 \\ 6 \end{bmatrix}
$$

- Choosing columns 4, 5, 6, 7, we have $B = I = B^{-1}$, so $\underline{x}_B = B^{-1}\underline{b} = \underline{b} \geq \underline{0}$, thus we obtain a basic feasible solution.
- Choosing columns 2, 5, 6, 7, we have

$$
B = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 3 & 0 & 0 & 1 \end{bmatrix}, B^{-1} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -3 & 0 & 0 & 1 \end{bmatrix}, x_B = \begin{bmatrix} 4 \\ 2 \\ 3 \\ -6 \end{bmatrix}
$$

and thus we obtain an infeasible basic solution.

**Number of basic feasible solutions**

At most one basic feasible solution for each choice of the $n - m$ non basic variables (to be set to zero) out of the $n$ variables:

$$\#\text{ basic feasible solutions } \leq \binom{n}{n-m} = \frac{n!}{(n-m)!(n-(n-m))!} = \binom{n}{m}.$$

# 3.4 Simplex method

Given any LP in standard form:

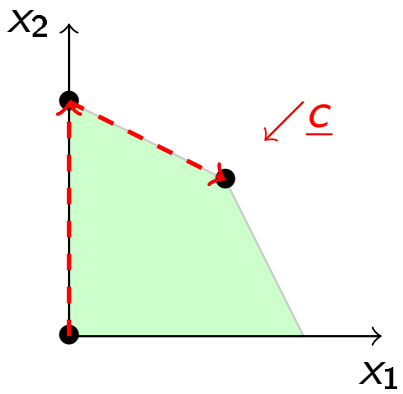
$$\begin{array}{ll} \min & z = \underline{c}^T \underline{x} \\ \text{s.t.} & A\underline{x} = \underline{b} \\ & \underline{x} \geq \underline{0} \end{array}$$

George Dantzig
(1914-2005)

Examine a sequence of basic feasible solutions with non-increasing objective function values until an optimal solution is reached or the LP is found to be unbounded.

At each iteration, move from a basic feasible solution to a "neighboring" basic feasible solution.

Generate a path (i.e., a sequence of adjacent vertices) along the edges of the polyhedron of the feasible solutions until an optimal vertex is reached.



Given the correspondence between the basic feasible solutions and the vertices, we need to describe how to:

- Find an initial vertex or establish that the LP is infeasible. (Achieved by applying the same method to another LP, see later)
- Determine whether the current vertex is optimal.
- Move from a current vertex to a better adjacent vertex (in terms of objective function value) or establish that the LP is unbounded.

# 3.4.1 Optimality test

Given a LP $\min\{\underline{c}^T\underline{x} : \ A\underline{x} = \underline{b}, \underline{x} \geq \underline{0}\}$ and a feasible basis $B$ of $A$, $A\underline{x} = \underline{b}$ can be rewritten as

$$B\underline{x}_B + N\underline{x}_N = \underline{b} \Rightarrow \underline{x}_B = B^{-1}\underline{b} - B^{-1}N\underline{x}_N$$

with $B^{-1}\underline{b} \geq 0$.

Then, a basic feasible solution is such that $\underline{x}_B = B^{-1}\underline{b}$, $\underline{x}_N = \underline{0}$.

By substitution we express the objective function in terms of only the non basic variables $\underline{x}_N$ (for the current basis $B$):

$$\begin{aligned}
\underline{c}^T\underline{x} &= (\underline{c}_B^T \ \underline{c}_N^T)\begin{bmatrix} \underline{x}_B \\ \underline{x}_N \end{bmatrix} = (\underline{c}_B^T \ \underline{c}_N^T)\begin{bmatrix} B^{-1}\underline{b} - B^{-1}N\underline{x}_N \\ \underline{x}_N \end{bmatrix} \\
&= \underline{c}_B^T B^{-1}\underline{b} - \underline{c}_B^T B^{-1} N\underline{x}_N + \underline{c}_N^T\underline{x}_N \\
&= \underline{c}_B^T B^{-1}\underline{b} + (\underline{c}_N^T - \underline{c}_B^T B^{-1} N)\underline{x}_N
\end{aligned}$$

$z_0 = \underline{c}_B^T B^{-1}\underline{b}$ is the cost of $\underline{x} = (\underline{x}_B \ \underline{x}_N)^T$, where $\underline{x}_B = B^{-1}\underline{b}$, $\underline{x}_N = \underline{0}$.

The reduced costs of the non basic variables are denoted by $\bar{\underline{c}}_N^T := \underline{c}_N^T - \underline{c}_B^T B^{-1} N$.

## Definition 17

$$\underline{\bar{c}}^T = \underline{c}^T - \underline{c}_B^T B^{-1} A = [\overbrace{\underline{c}_B^T - \underline{c}_B^T B^{-1} B}^{\underline{0}^T}, \overbrace{\underline{c}_N^T - \underline{c}_B^T B^{-1} N}^{\underline{\bar{c}}_N^T}]$$

is the vector of reduced costs with respect to the basis $B$.

- The reduced costs are defined also for basic variables, but $\underline{\bar{c}}_B = \underline{0}$.
- $\bar{c}_j$ represents the change in the objective function value if non basic $x_j$ would be increased from 0 to 1 while keeping all other non basic variables to 0.
- The solution value changes by $\Delta z = \bar{c}_j \Delta x_j$.

Consider a LP $\min\{\underline{c}^T\underline{x} : \ A\underline{x} = \underline{b}, \underline{x} \geq \underline{0}\}$ and a feasible basis $B$.

## Proposition 1

*If $\bar{\underline{c}}_N \geq \underline{0}$ then the basic feasible solution $(\underline{x}_B^T, \underline{x}_N^T)$, where $\underline{x}_B = B^{-1}\underline{b} \geq \underline{0}$ and $\underline{x}_N = \underline{0}$, of cost $\underline{c}_B^T B^{-1}\underline{b}$ is a global optimum.*

## Proof.

$\bar{\underline{c}}^T \geq \underline{0}^T$ implies that

$$\underline{c}^T\underline{x} = \underline{c}_B^T B^{-1}\underline{b} + \bar{\underline{c}}_N^T \underline{x}_N \geq \underline{c}_B^T B^{-1}\underline{b} \qquad \forall \underline{x} \geq \underline{0}, A\underline{x} = \underline{b}.$$

□

**Note:**

- For maximization problems we check whether $\bar{\underline{c}}_N \leq \underline{0}$.
- This optimality condition is sufficient but in general not necessary.

**Example**

$$\begin{array}{lll} \min & z = -x_1 - x_2 & \\ \text{s. t.} & x_1 - x_2 + s_1 = 1 & \\ & x_1 + x_2 + s_2 = 3 & \\ & x_1,\ x_2,\ s_1,\ s_2 \geq 0 & \end{array} \qquad A = \begin{bmatrix} 1 & -1 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{bmatrix} \qquad \underline{c} = [-1\ \ -1\ 0\ 0]^T$$

Consider basic feasible solution: $\underline{x}_B = (x_1, s_2)^T = (1,2)^T$
$\underline{x}_N = (x_2, s_1)^T = \underline{0}^T$ with $z = -1$ and

$$B = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix},\ N = \begin{bmatrix} -1 & 1 \\ 1 & 0 \end{bmatrix}$$

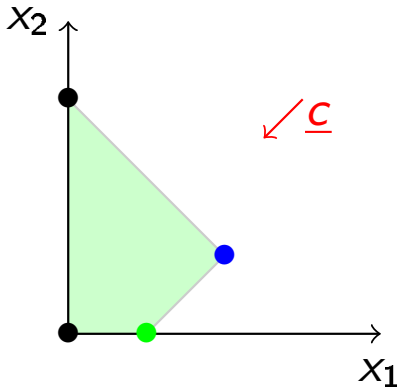
Therefore

$$\underline{c}_B^T = \underline{c}_N^T = [-1\ 0]^T,\ B^{-1} = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}$$

and the reduced costs for $x_2, s_1$ are

$$\bar{\underline{c}}_N^T = \underline{c}_N^T - \underline{c}_B^T B^{-1} N = [-2\ 0]^T$$

Since $\bar{c}_2 = -2 < 0$, if $x_2$ can be increased from 0 to 1 while keeping $s_1 = 0$, then $z$ can vary by $-2$.

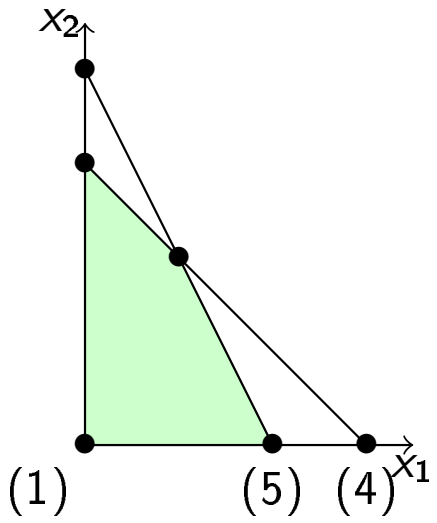



Increasing $x_2$ from 0 to 1 keeping $s_1 = 0$, we move (along the edge) from the green vertex $(1,0)^T$ with $z = -1$ to the blue vertex $(2,1)^T$ with $z = -3$.

# 3.4.2 Move to an adjacent vertex (basic feasible solution)

**Example**

$$\begin{aligned} x_1 + x_2 + s_1 \quad &= 6 \\ 2x_1 + x_2 \quad + s_2 &= 8 \\ x_1,\ x_2,\ s_1,\ s_2 &\geq 0 \end{aligned}$$




Move from vertex (1) to vertex (5):

- In (1) $x_1 = 0, x_2 = 0 \Rightarrow s_1 = 6, s_2 = 8$ with $\underline{x}_B = (s_1, s_2)^T$, $\underline{x}_N = (x_1, x_2)T$
- In (5) $x_2 = 0, s_2 = 0 \Rightarrow x_1 = 4, s_1 = 2$ with $\underline{x}_B = (x_1, s_1)^T$, $\underline{x}_N = (x_2, s_2)^T$

Thus, $x_1$ enters the basis $B$ and $s_2$ exits the basis $B$.

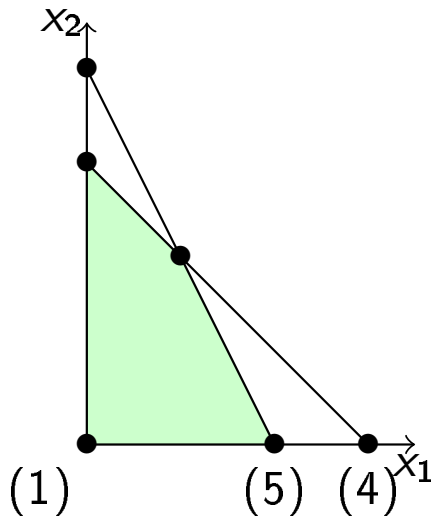
**Note:** When moving from the current vertex to an adjacent vertex, we substitute one column of $B$ (that of $s_2$) with one column of $N$ (that of $x_1$).

By expressing the basic variables in terms of the non basic variables, we obtain

$$s_1 = 6 - x_1 - x_2$$

$$s_2 = 8 - 2x_1 - x_2$$

Now we increase $x_1$, while keeping $x_2 = 0$.




Since $s_1 = 6 - x_1 \geq 0$ implies $x_1 \leq 6$ and $s_2 = 8 - 2x_1 \geq 0$ implies $x_1 \leq 8/2 = 4$, the upper limit on the increase of $x_1$ is: $x_1 \leq \min\{6, 4\} = 4$.

We move from vertex (1) to vertex (5) by letting $x_1$ enter the basis and $s_2$ exit the basis ($s_1 = 2$ and $s_2 = 0$).

**Note:** When $x_1 = 6$, we obtain the infeasible basic solution (4).

## General case

Given a basis $B$, the system

$$A\underline{x} = \underline{b} \Leftrightarrow \sum_{j=1}^{n} a_{ij}x_j = b_i, \text{ for } i = 1, \ldots, m$$

can be expressed in canonical form

$$\underline{x}_B + B^{-1}N\underline{x}_N = B^{-1}\underline{b} \Leftrightarrow \underline{x}_B + \bar{N}\underline{x}_N = \bar{\underline{b}}$$

which emphasizes the basic feasible solution $(\underline{x}_B, \underline{x}_N) = (B^{-1}\underline{b}, \underline{0})$.
This amounts to pre-multiply the system by $B^{-1}$:

$$\overbrace{B^{-1}B}^{=I}\,\underline{x}_B + \overbrace{B^{-1}N}^{=\bar{N}}\,\underline{x}_N = \overbrace{B^{-1}\underline{b}}^{=\bar{\underline{b}}}$$

In the canonical form

$$x_{B_i} + \sum_{j=1}^{n-m} \bar{a}_{ij} x_{N_j} = \bar{b}_i \quad \text{for } i = 1, \ldots, m \quad (I\underline{x}_B + \bar{N}\underline{x}_N = \underline{\bar{b}})$$

basic variables are expressed in terms of non basic variables: $\underline{x}_B = \underline{\bar{b}} - \bar{N}\underline{x}_N$.

If we increase the value of a non basic $x_s$ (from 0) while keeping all other non basic variables to 0, the system becomes

$$x_{B_i} + \bar{a}_{is} x_s = \bar{b}_i \Leftrightarrow x_{B_i} = \bar{b}_i - \bar{a}_{is} x_s \quad \text{for } i = 1, \ldots, m$$

To guarantee $x_{B_i} \geq 0$ for each $i$, we need to satisfy

$$\bar{b}_i - \bar{a}_{is} x_s \geq 0 \Rightarrow x_s \leq \bar{b}_i / \bar{a}_{is} \quad \text{for } \bar{a}_{is} \geq 0.$$

The value of $x_s$ can be increased up to

$$\theta^* = \min_{i=1,\ldots,m} \left\{ \bar{b}_i / \bar{a}_{is} : \bar{a}_{is} \geq 0 \right\}.$$

(If $\bar{a}_{is} \leq 0$ for all $i$, there is no limit to the increase of $x_s$.) The value of $x_r$ where

$$r = \text{argmin}_{i=1,\ldots,m} \left\{ \bar{b}_i / \bar{a}_{is} : \bar{a}_{is} \geq 0 \right\}$$

decreases to 0 and exits the basis.

## Change of basis (for minimization LP)

Let $B$ be a feasible basis and $x_s$ (in $\underline{x}_N$) a non basic with $\bar{c}_s < 0$.

- Increase $x_s$ as much as possible ($x_s$ "enters the basis") while keeping the other non basic variables equal to 0.
- A basic variable $x_r$ (in $\underline{x}_B$) such that $x_r \geq 0$ imposes the tightest upper bound $\theta^*$ on the increase of $x_s$, decreases to 0 ($x_r$ "exits the basis").

If $\theta^* > 0$, the new BFS has a better objective function value.

The new basis differs w.r.t. the previous one by a single column (adjacent vertices).

To go from the canonical form for current BFS

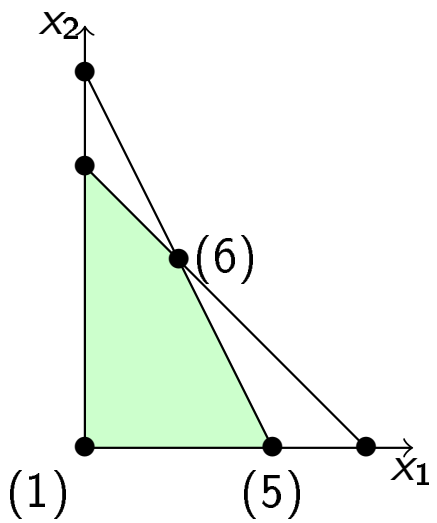
$$B^{-1}B\underline{x}_B + B^{-1}N\underline{x}_N = B^{-1}\underline{b}$$

to that of an adjacent one, no need to compute $B^{-1}$ from scratch.

$B^{-1}$ of the new basis $B$ can be obtained by applying to the inverse of the previous basis (difference: a single column) a unique "pivoting" operation.

## Example

$$\begin{array}{lrl} \min & -x_1 - x_2 & \\ \text{s. t.} & x_1 + x_2 + s_1 & = 6 \\ & 2x_1 + x_2 \quad + s_2 & = 8 \\ & x_1,\ x_2,\ s_1,\ s_2 & \geq 0 \end{array}$$




Move from vertex (1) to vertex (5).

System in canonical form w.r.t. the basis with $s_1$ and $s_2$ basic (vertex (1)):

$$\begin{array}{lcl} s_1 & = & 6 - x_1 - x_2 \\ s_2 & = & 8 - 2x_1 - x_2 \end{array} \qquad \begin{array}{rrrrr} x_1 & +x_2 & +s_1 & & = 6 \\ 2x_1 & +x_2 & & +s_2 & = 8 \end{array}$$

pivot

columns of basic variables

$x_1$ enters the basis and $s_2$ exits

System in canonical form w.r.t. the basis with $x_1$ and $s_1$ basic (vertex (5)):

$$\begin{array}{c} (4 - \frac{1}{2}x_2 - \frac{1}{2}s_2) + x_2 + s_1 = 6 \\ x_1 = 4 - \frac{1}{2}x_2 - \frac{1}{2}s_2 \end{array} \quad \Rightarrow \quad \begin{array}{rrrr} \frac{1}{2}x_2 & +s_1 & -\frac{1}{2}s_2 & = 2 \\ x_1 \quad +\frac{1}{2}x_2 & & +\frac{1}{2}s_2 & = 4 \end{array}$$

## "Pivoting" operation

Same operations used in the Gaussian elimination method to solve systems of linear equations.

Given $A\underline{x} = \underline{b}$:

1. Select a coefficient $\bar{a}_{rs} \neq 0$ (the "pivot)"
2. Divide the $r$-th row by $\bar{a}_{rs}$.
3. For each row $i$ with $i \neq r$ and $\bar{a}_{is} \neq 0$, subtract the resulting $r$-th row multiplied by $\bar{a}_{is}$.

$$
\begin{array}{c|cccc|c}
 & s & & A & & \underline{b} \\
\hline
 & 1 & 1 & 1 & 0 & 6 \\
r \rightarrow & 2 & 1 & 0 & 1 & 8 \\
\hline
\end{array}
\longrightarrow
\begin{array}{|cccc|c|}
\hline
0 & 1/2 & 1 & -1/2 & 2 \\
1 & 1/2 & 0 & 1/2 & 4 \\
\hline
\end{array}
$$

pivot
columns of basic variables

**Note**: This does not affect the feasible solutions.

## Moving to an adjacent basic feasible solution (vertex)

Goals: - improve the objective function value
- preserve feasibility

1. Which non basic variable enters the basis? (choice of pivot column $s$.)
   - Any one with reduced cost $\bar{c}_j < 0$.
   - One that yields the maximum $\Delta z$ w.r.t. $z = c_B^T B^{-1} b$ (the actual decrement $\Delta z$ also depends on $\theta^*$).
   - Bland's rule: $s = \min\{j : \bar{c}_j < 0\}$. ($\bar{c}_j > 0$ for maximization problems.)

2. Which basic variable leaves the basis?

   **Min ratio test**: index $i$ with smallest $\bar{b}_i/\bar{a}_{is} = \theta^*$ (thightest upper bound on increase of $x_s$) among those with $\bar{a}_{is} > 0$ (otherwise no limit!).
   - Bland's rule: $r = \min\{i : \bar{b}_i/\bar{a}_{is} = \theta^*, \bar{a}_{is} > 0\}$
   - randomly...

**Unboundedness**: If $\exists \bar{c}_j < 0$ with $\bar{a}_{ij} \leq 0 \forall i$, no element of the $j$-th column can play the role of a pivot. $\Rightarrow$ The minimization problem is unbounded!

# 3.4.3 "Tableau" representation

System $\begin{cases} z = \underline{c}^T \underline{x} \\ A\underline{x} = \underline{b} \end{cases}$ with (implicit) nonnegativity constraints

Initial tableau:

| | $x_1$ | $\cdots$ | $x_n$ | |
|---|---|---|---|---|
| 0 | | $\underline{c}^T$ | | $\leftarrow$ objective function |
| $\underline{b}$ | | $A$ | | $\leftarrow$ $m$ rows |

The leftmost column (0-th column) contains:

- right hand side of the objective function ($0 = \underline{c}^T \underline{x} - z$)
- right hand side vector

Considering a basis $B$ and a partition $A = [B\ N]$, with $0 = \underline{c}_B^T \underline{x}_B + \underline{c}_N^T \underline{x}_N - z$, we have:

| | $x_1$ | $\cdots$ | $x_m$ | $x_{m+1}$ | $\cdots$ | $x_n$ | $z$ |
|---|---|---|---|---|---|---|---|
| $0$ | | $\underline{c}_B^T$ | | | $\underline{c}_N^T$ | | $-1$ |
| $\underline{b}$ | | $B$ | | | $N$ | | $0$ $\vdots$ $0$ |

the column of $z$ is omitted

By "pivoting" operations (or pre-multiplying by $B^{-1}$) we put the tableau in canonical form with respect to $B$:

| | | $x_1$ | $\cdots$ | $x_m$ | $x_{m+1}$ | $\cdots$ | $x_n$ |
|---|---|---|---|---|---|---|---|
| $-z$ | $-z_0$ | $0$ | $\cdots$ | $0$ | | $\bar{\underline{c}}_N^T$ | |
| $x_{B[1]}$ $\vdots$ $x_{B[m]}$ | $\bar{\underline{b}}$ | | $I$ | | | $\bar{N}$ | |
| | | basic variables | | | non basic variables | | |

$$z = z_0 + \bar{\underline{c}}_N^T \underline{x}_N$$

$$z_0 = \underline{c}_B^T B^{-1} \underline{b}$$

$$\bar{\underline{b}} = B^{-1} \underline{b}$$

$$\bar{N} = B^{-1} N$$

**Example**

$$\begin{array}{lrl} \min & -x_1 \ -x_2 & \\ \text{s. t.} & 6x_1+4x_2+x_3 & = 24 \\ & 3x_1-2x_2 \quad +x_4 & = 6 \\ & x_1, \ x_2, \ x_3, \ x_4 & \geq 0 \end{array}$$

Tableau w.r.t. the basis with columns 3, 4:

| | | $x_1$ | $x_2$ | $x_3$ | $x_4$ |
|---|---|---|---|---|---|
| $-z$ | 0 | $-1$ | $-1$ | 0 | 0 |
| $x_3$ | 24 | 6 | 4 | 1 | 0 |
| $x_4$ | 6 | 3 | $-2$ | 0 | 1 |

pivot basic variables

Pivot w.r.t. 3 amounts to deriving an expression for $x_1$ from the pivot row and substituting it in all other rows.

$x_1$ enters in the basis and $x_4$ exits the basis

Tableau w.r.t. the new basis with columns 1, 3:

| | | $x_1$ | $x_2$ | $x_3$ | $x_4$ |
|---|---|---|---|---|---|
| $-z$ | 2 | 0 | $-5/3$ | 0 | $1/3$ |
| $x_3$ | 12 | 0 | 8 | 1 | $-2$ |
| $x_1$ | 2 | 1 | $-2/3$ | 0 | $1/3$ |

Corresponding basic feasible solution: $\underline{x} = (2, 0, 12, 0)^T$, with $z = -2$.

$x_2$ is the only non basic variable that can "enter" the basis ($\bar{c}_2 = -5/3 < 0$)
$x_3$ is the only basic variable that can "exit" the basis ($\bar{a}_{rs} = 8 > 0$)

| | | $x_1$ | $x_2$ | $x_3$ | $x_4$ |
|---|---|---|---|---|---|
| $-z$ | 2 | 0 | $-5/3$ | 0 | $1/3$ |
| $x_3$ | 12 | 0 | 8 | 1 | $-2$ |
| $x_1$ | 2 | 1 | $-2/3$ | 0 | $1/3$ |

| | | $x_1$ | $x_2$ | $x_3$ | $x_4$ |
|---|---|---|---|---|---|
| $-z$ | $9/2$ | 0 | 0 | $5/24$ | $-1/12$ |
| $x_2$ | $3/2$ | 0 | 1 | $1/8$ | $-1/4$ |
| $x_1$ | 3 | 1 | 0 | $1/12$ | $\frac{1}{6}$ |

| | | $x_1$ | $x_2$ | $x_3$ | $x_4$ |
|---|---|---|---|---|---|
| $-z$ | 6 | $1/2$ | 0 | $1/4$ | 0 |
| $x_2$ | 6 | $3/2$ | 1 | $1/4$ | 0 |
| $x_4$ | 18 | 6 | 0 | $1/2$ | 1 |

Since all reduced costs $\geq 0$, the basic feasible solution $\underline{x}^* = (0, 6, 0, 18)^T$ with $z^* = -6$ is optimal.

**Algorithm 1:** Simplex algorithm (LP with minimization)

Let $B[1], \ldots, B[m]$ be the column indices of the initial feasible basis $B$.
Construct the initial tableau $\bar{A} = \{\bar{a}[i,j] : 0 \leq i \leq m, 0 \leq j \leq n\}$ in canonical form w.r.t. $B$.
*unbounded* $\leftarrow$ false
*optimal* $\leftarrow$ false
**while** *optimal* $=$ *false* ***and*** *unbounded* $=$ *false* **do**
  **if** $\bar{a}[0,j] \geq 0 \forall j = 1, \ldots, n$ **then** *optimal* $\leftarrow$ true`/* for LP with min */`

  **else**
    Select a non basic variable $x_s$ with $\bar{a}[0,s] < 0$ `/* negative reduced cost */`

    **if** $\bar{a}[i,s] \leq 0 \forall i = 1, \ldots, m$ **then** *unbounded* $\leftarrow$ true
    **else**
      $r \leftarrow \text{argmin} \left\{ \frac{\bar{a}[i,0]}{\bar{a}[i,s]} \text{ with } 1 \leq i \leq m \text{ and } \bar{a}[i,s] > 0 \right\}$
      *pivot*$(r,s)$ `/* update tableau */`
      $B[r] \leftarrow s$

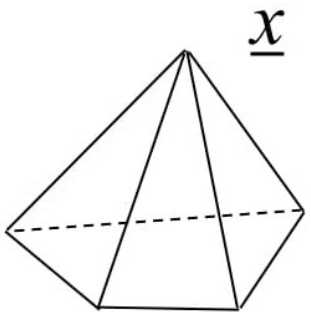
# 3.4.4 Degenerate basic feasible solutions and convergence

## Definition 18

A basic feasible solution $\underline{x}$ is degenerate if it contains at least one basic variable $x_j = 0$.

A solution $\underline{x}$ with more than $n - m$ zeroes corresponds to several distinct bases!

Same vertex:




More than $n$ constraints (the $m$ of $A\underline{x} = \underline{b}$ and more than $n - m$ among the $n$ of $\underline{x} \geq \underline{0}$) are satisfied with equality ("active").

In the presence of degenerate basic feasible solutions (BFSs), a basis change may not decrease the objective function value:

if the current BFS is degenerate, one can have $\theta^* = 0$ and hence the new BFS is identical (same vertex).

Note that a degenerate BFS can arise from a non degenerate one: even if $\theta^* > 0$, several basic variables may go to 0 when $x_s$ is increased to $\theta^*$.

One can cycle through a sequence of "degenerate" bases associated to the same vertex.

Several "anticycling" rules have been proposed for the choice of the variables that enter and exit the bases (indices $r$ and $s$).

**Bland's rule:** Among all candidate variables to enter/exit the basis ($x_s/x_r$) always select the one with smallest index.

## Proposition 2

*The Simplex algorithm with Bland's rule terminates after* $\leq \binom{n}{m}$ *iterations.*

**Note:** The number of pivots is finite.

In some "pathological" cases (see e.g. Klee & Minty 72), the number of iterations grows exponentially w.r.t. $n$ or $m$.

However, the Simplex algorithm is overall very efficient:

Extensive experimental campaigns show that number of iterations grows linearly w.r.t. $m$ ($m \leq \ldots \leq 3m$) and very slowly ($\approx$ logarithmically) w.r.t. $n$.

# 3.4.5 Two-phase simplex method

Phase I: Determine an initial basic feasible solution.

**Example:**

$$\begin{array}{lrrrl} \min & x_1 & & +x_3 & \\ \text{s.t.} & x_1 & +2x_2 & & \leq 5 \\ & & x_2 & +2x_3 & = 6 \\ & x_1, & x_2, & x_3 & \geq 0 \end{array} \quad \Rightarrow \quad \begin{array}{lrrrrl} \min & x_1 & & +x_3 & & \\ \text{s.t.} & x_1 & +2x_2 & & +x_4 & = 5 \\ & & x_2 & +2x_3 & & = 6 \\ & x_1, & x_2, & x_3, & x_4 & \geq 0 \end{array}$$

There is no sub-matrix $I_{2\times 2}$ of $A$!

Given problem $(P)$: $\begin{array}{ll} \min & z = \underline{c}^T \underline{x} \\ \text{s.t.} & A\underline{x} = \underline{b} \\ & \underline{x} \geq \underline{0} \end{array}$ with $\underline{b} \geq \underline{0}$, the auxiliary LP with artificial variables $y_i$, $i = 1, \ldots, m$, is:

$$(P_A) \quad \begin{array}{ll} \min & v = \sum_{i=1}^m y_i \\ \text{s.t.} & A\underline{x} + I\underline{y} = \underline{b} \\ & \underline{x} \geq \underline{0}, \underline{y} \geq \underline{0} \end{array}$$

Obvious initial basic feasible solution: $\underline{y} = \underline{b} \geq \underline{0}$ and $\underline{x} = \underline{0}$.

1. If $v^* > 0$, then (P) is infeasible.
2. If $v^* = 0$, clearly $\underline{y}^* = 0$ and $\underline{x}^*$ is a basic feasible solution of (P).
   - If $y_i$ is non basic $\forall i$, with $1 \leq i \leq m$, delete the corresponding columns and obtain a tableau in canonical form w.r.t. a basis; the row of $z$ must be determined by substitution.
   - If there is a basic $y_i$ (the BFS is degenerate), we perform a "pivot" operation w.r.t. a coefficient $\neq 0$ of the row of $y_i$ so as to "exchange" $y_i$ with any non basic variable $x_j$.

cf. example

**Example:**

$$(P)\quad \begin{array}{lrl} \min & x_1+x_2+10x_3 & \\ \text{s.t.} & x_2+4x_3 & = 2 \\ & -2x_1+x_2-6x_3 & = 2 \\ & x_1,\ x_2,\ x_3 & \geq 0 \end{array} \qquad (P_A)\quad \begin{array}{lrl} \min & v = y_1+y_2 & \\ \text{s.t.} & x_2+4x_3+y_1 & = 2 \\ & -2x_1+x_2-6x_3+y_2 & = 2 \\ & x_1,\ x_2,\ x_3,\ y_1,\ y_2 & \geq 0 \end{array}$$

By expressing $y_1$ and $y_2$ in terms of $x_1$, $x_2$ and $x_3$, we put $v = y_1 + y_2$ in canonical form: $v = 4 + 2x_1 - 2x_2 + 2x_3$.

| | | $x_1$ | $x_2$ | $x_3$ | $y_1$ | $y_2$ |
|---|---|---|---|---|---|---|
| $-v$ | $-4$ | 2 | $-2$ | 2 | 0 | 0 |
| $y_1$ | 2 | 0 | 1 | 4 | 1 | 0 |
| $y_2$ | 2 | $-2$ | 1 | $-6$ | 0 | 1 |

| | | $x_1$ | $x_2$ | $x_3$ | $y_1$ | $y_2$ |
|---|---|---|---|---|---|---|
| $-v$ | 0 | 2 | 0 | 10 | 2 | 0 |
| $x_2$ | 2 | 0 | 1 | 4 | 1 | 0 |
| $y_2$ | 0 | $-2$ | 0 | $-10$ | $-1$ | 1 |

By swapping the roles of $y_2$ and $x_1$ (or $x_3$), we identify an optimal solution of the Phase I problem: $\underline{x}^* = (0, 2, 0)^T$, $\underline{y}^* = (0, 0)^T$, with optimal value $v^* = 0$.

By selecting as "pivot" the coefficient $-2$ of the row of $y_2$, we obtain:

| | | $x_1$ | $x_2$ | $x_3$ | $y_1$ | $y_2$ |
|---|---|---|---|---|---|---|
| $-v$ | 0 | 0 | 0 | 0 | 1 | 1 |
| $x_2$ | 2 | 0 | 1 | 4 | 1 | 0 |
| $x_1$ | 0 | 1 | 0 | 5 | 1/2 | $-1/2$ |

corresponds to same optimal BFS of $(P_A)$ and $\underline{x} = (0, 2, 0)^T$ is a BFS of $(P)$

The original LP with

$$z = x_1 + x_2 + 10x_3$$

is not in canonical form since $x_1$ and $x_2$ are basic variables ($x_3$ is non basic).

By substituting

$$\begin{cases} x_2 = 2 - 4x_3 \\ x_1 = -5x_3, \end{cases}$$

we obtain $z = 2 + x_3$ and the tableau corresponding to the initial BFS of $(P)$:

| | | $x_1$ | $x_2$ | $x_3$ |
|---|---|---|---|---|
| $-v$ | -2 | 0 | 0 | 1 |
| $x_2$ | 2 | 0 | 1 | 4 |
| $x_1$ | 0 | 1 | 0 | 5 |

Since $\overline{c}_3 > 0$, the BFS found is (already) optimal and no need for Phase II!

# 3.4.6 Polynomial-time algorithms for LP

- Ellipsoid method (L. Khachiyan 1979)

  Theoretically important.

- Interior point methods (N. Karmarkar 1984, . . . )

Narendra Karmarkar (1957-)

Very efficient variants (e.g. barrier methods) for some types of instances (e.g. sparse and large-scale).